form. This helps the PDP focus on evaluating attributes rather than determining their origins.

5. **Let the PIP and PEP handle credential validity concerns.** Revocation checks, issuer validations, and signature verification belong in the enforcement pipeline, not in policy logic. Policies evaluate attributes, not cryptographic proofs. Note, for example, that the policy shown in the preceding section did not reference details like whether credentials were revoked or expired.

By centralizing issuer trust and credential interpretation in the PIP and PEP, ACME keeps policies simple, strong, and protected from ecosystem changes. This enables ACME to increase the number and types of credential issuers it collaborates with while keeping its authorization logic both stable and easy to understand.

16.7 Protocols and ecosystem integration

The mechanics of verifiable credentials involve more than just the credential itself. A functioning ecosystem relies on three interconnected layers: issuance protocols that describe how credentials are created and delivered to holders, presentation protocols that control how holders respond to attribute requests from verifiers, and wallet standards that specify how credentials are stored, protected, and managed on the holder's device. These layers differ in how they handle discovery, transport, and user interaction, but they all support the same authorization pattern shown in figure 16.3: The PEP requests attributes, the holder assembles a presentation, and the PDP evaluates normalized claims that the PEP has already verified.

Two protocol families dominate current deployments. The first follows the W3C Verifiable Credentials Data Model (https://www.w3.org/TR/vc-data-model-2.0/), where issuers use DID-based identifiers and sign credentials that wallets store. Holders respond directly to a verifier's request with a signed presentation that the verifier checks using issuer keys discovered through DID resolution to discover information about the DID identifier. The second family builds on existing web identity infrastructure through OpenID for Verifiable Credentials (https://openid.net/sg/openid4vc/),